

GRIDGUARD AI 4.0

National Smart Grid Executive Intelligence Report

Generated: 2026-02-07T22:37:01.210Z
Scope: National operational risk, theft intelligence, transformer reliability, alert posture, and recommended actions.
Source: Live platform data (regions, alerts, transformer telemetry, meter anomalies, and reading streams).

Executive Narrative

This executive brief presents a formal live operational assessment of GRIDGUARD AI 4.0 based on current telemetry, anomaly outputs, and alert intelligence.

At the time of generation, national average theft score stands at 22.37% and national distribution loss stands at 7.52%. The transformer risk index is 42.2% and the carbon optimization score is 70.43%.

System footprint currently includes 35 regions, 120 transformers, and 1080 smart meters.

Risk zoning currently indicates 1 red zones, 0 yellow zones, and 34 green zones.

The estimated daily financial impact of loss is INR 34,507.23 based on recent load behavior and live loss conditions.

Executive Summary: National loss elevated, with concentrated theft in high-density regions. Transformer risk moderate. Recommend targeted inspections, load balancing, and proactive maintenance.

Board Note: Formal monthly review

Key Metrics

National Theft Score (Avg): 22.37%
National Distribution Loss (Avg): 7.52%
Transformer Risk (Avg): 42.2%
Carbon Optimization (Avg): 70.43%
Open Alerts: 38
High Alerts (24h): 13
Estimated Daily Financial Impact (INR): 34,507.23

National Operational Snapshot

Total Regions Monitored: 35
Total Transformers Monitored: 120
Total Smart Meters Monitored: 1080
National Theft Score (Avg): 22.37%
National Distribution Loss (Avg): 7.52%
National Transformer Risk (Avg): 42.2%
National Carbon Optimization Score (Avg): 70.43%

Risk Zone Distribution

Red Zones (>70 theft score): 1
Yellow Zones (40-70 theft score): 0
Green Zones (<40 theft score): 34

Alert Intelligence

Total Alerts Logged: 44
Open Alerts (Unacknowledged): 38
High Severity Alerts in Last 24h: 13

- [2026-02-07T22:36:28.440Z] (HIGH) Gujarat: Anomaly detected for meter 375a665d-c076-48e8-bb0c-40aebd555993
- [2026-02-07T22:36:04.299Z] (HIGH) Tamil Nadu: Anomaly detected for meter 98223e50-2a80-40ab-94a6-dc755380df6e
- [2026-02-07T22:35:22.046Z] (HIGH) Bihar: Anomaly detected for meter b975f589-7d2e-4122-aa43-4b32664000ed
- [2026-02-07T22:34:48.808Z] (MEDIUM) Maharashtra: Anomaly detected for meter 9fcabb1c-bcf1-4239-a3a2-b1268be17037
- [2026-02-07T22:34:45.783Z] (MEDIUM) Karnataka: Anomaly detected for meter e1724502-83e6-4776-b9dd-2ab76bca9b93
- [2026-02-07T22:28:46.445Z] (MEDIUM) Chandigarh: Anomaly detected for meter 9651513e-6537-4b11-8e61-42fb0f7d6b28
- [2026-02-07T22:08:09.430Z] (HIGH) Delhi: Anomaly detected for meter c3329c11-04b1-49bb-b1fe-1ce3edef795b
- [2026-02-07T22:07:24.212Z] (MEDIUM) Karnataka: Anomaly detected for meter fad79635-0dde-47ff-9aee-d985c1d2009e
- [2026-02-07T22:06:26.490Z] (MEDIUM) Gujarat: Anomaly detected for meter 6aa1eb52-2f44-4772-bc90-d2bcb05b784a
- [2026-02-07T22:05:14.043Z] (MEDIUM) Karnataka: Anomaly detected for meter b036bc68-d33c-4d1e-9a99-b1855888bedd
- [2026-02-07T22:04:25.807Z] (HIGH) Maharashtra: Anomaly detected for meter a0685f8a-8e7f-4da5-ac14-ea779ee7a290
- [2026-02-07T22:04:10.747Z] (MEDIUM) Gujarat: Anomaly detected for meter d415786e-9751-4774-9877-563af57fdd7b

Top High-Risk Regions

- 1. Bihar | Theft 74% | Loss 24.81% | Transformer Risk 60.25% | Carbon 70% | Updated 2026-02-07T22:35:22.045Z
- 2. Mizoram | Theft 34% | Loss 11.29% | Transformer Risk 43.27% | Carbon 59% | Updated 2026-02-07T22:17:27.901Z
- 3. West Bengal | Theft 34% | Loss 6.71% | Transformer Risk 41.29% | Carbon 73% | Updated 2026-02-07T22:17:27.901Z
- 4. Sikkim | Theft 33% | Loss 5.17% | Transformer Risk 48.68% | Carbon 85% | Updated 2026-02-07T22:17:27.901Z
- 5. Himachal Pradesh | Theft 32% | Loss 9.67% | Transformer Risk 27.63% | Carbon 64% | Updated 2026-02-07T22:17:27.901Z
- 6. Dadra and Nagar Haveli | Theft 30% | Loss 4.91% | Transformer Risk 23.51% | Carbon 62% | Updated 2026-02-07T22:17:27.901Z
- 7. Jharkhand | Theft 29% | Loss 6.51% | Transformer Risk 45.79% | Carbon 67% | Updated 2026-02-07T22:17:27.901Z

- 8. Andaman and Nicobar | Theft 27% | Loss 8.9% | Transformer Risk 53.7% | Carbon 70% | Updated 2026-02-07T22:35:58.260Z
- 9. Madhya Pradesh | Theft 26% | Loss 10.36% | Transformer Risk 34.43% | Carbon 70% | Updated 2026-02-07T22:17:27.901Z
- 10. Orissa | Theft 25% | Loss 7.43% | Transformer Risk 63.02% | Carbon 67% | Updated 2026-02-07T22:17:27.901Z

Top Vulnerable Transformers

- 1. Karnataka | Transformer 5a0a5c5a | Load 74.39% | Temp 47.1 C | Health 0.38 | Risk GREEN
- 2. Delhi | Transformer 67102a47 | Load 74.06% | Temp 66.74 C | Health 0.38 | Risk GREEN
- 3. Karnataka | Transformer bf18b616 | Load 74.02% | Temp 58.48 C | Health 0.38 | Risk GREEN
- 4. Assam | Transformer 6d23a185 | Load 72.69% | Temp 63.73 C | Health 0.39 | Risk GREEN
- 5. Bihar | Transformer bb9efdb5 | Load 71.38% | Temp 62.16 C | Health 0.41 | Risk GREEN
- 6. Gujarat | Transformer 52a08580 | Load 71.21% | Temp 63.2 C | Health 0.41 | Risk GREEN
- 7. Gujarat | Transformer c7633930 | Load 70.55% | Temp 62.11 C | Health 0.41 | Risk GREEN
- 8. Delhi | Transformer d232ecfb | Load 68.41% | Temp 69.3 C | Health 0.43 | Risk GREEN
- 9. Tamil Nadu | Transformer f1486e92 | Load 68.09% | Temp 63 C | Health 0.43 | Risk RED
- 10. Andhra Pradesh | Transformer 9ab80d8c | Load 67.76% | Temp 57.37 C | Health 0.44 | Risk GREEN

Top Meter Anomalies (7-day window)

- 1. Delhi | Meter 9657afa8-a5b | Theft Probability 83% | Class RED | Detected 2026-02-07T21:50:38.095Z
- 2. Gujarat | Meter 375a665d-c07 | Theft Probability 82% | Class RED | Detected 2026-02-07T22:36:28.438Z
- 3. Tamil Nadu | Meter 98223e50-2a8 | Theft Probability 81% | Class RED | Detected 2026-02-07T22:36:04.297Z
- 4. Delhi | Meter c3329c11-04b | Theft Probability 79% | Class RED | Detected 2026-02-07T22:08:09.428Z
- 5. Tamil Nadu | Meter 44e5eb02-9fd | Theft Probability 79% | Class RED | Detected 2026-02-07T22:00:21.340Z
- 6. Gujarat | Meter c2ed3b5d-83d | Theft Probability 78% | Class RED | Detected 2026-02-07T21:51:44.434Z
- 7. Tamil Nadu | Meter aff96e62-56f | Theft Probability 77% | Class RED | Detected 2026-02-07T22:02:49.247Z
- 8. Maharashtra | Meter 532ca612-b10 | Theft Probability 76% | Class RED | Detected 2026-02-07T22:02:40.192Z
- 9. Bihar | Meter b975f589-7d2 | Theft Probability 74% | Class RED | Detected 2026-02-07T22:35:22.044Z
- 10. Delhi | Meter 7cfa8051-267 | Theft Probability 74% | Class RED | Detected 2026-02-07T22:01:45.874Z

24-Hour Load Trend

Average Power (24h): 2.528 kW

Estimated Daily Energy Loss: 4929.6 kWh

- 2026-02-07T21:30:00.000Z | Avg Power 2.807 kW | Readings 781
- 2026-02-07T22:30:00.000Z | Avg Power 2.249 kW | Readings 45

Recommended Actions

- Maintain active anomaly surveillance and sustain monthly anti-theft audit cadence in medium-risk states.
- Current national loss is within an acceptable band; continue preventive maintenance and loss monitoring.
- Initiate a 14-day transformer reliability sprint focused on high-load assets with low health index.
- Establish an alert triage war-room with SLA-based acknowledgement and closure tracking.
- Use LLM assistant for weekly executive briefings, including state-wise risk rationale and recommended field actions.

Methodology Note

This report is generated from live PostgreSQL operational data and near-real-time simulator/ingest streams.

Risk zone classification follows GRIDGUARD thresholds based on theft probability and transformer stress signals.

Financial impact estimation uses current average power, meter coverage, and loss percentage with an indicative INR conversion factor.

Generated by GRIDGUARD AI 4.0 automated reporting engine.